



JKBOSE 2026

CLASS 11TH

LINEAR INEQUALITIES

MATHS

LECTURE-01

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TOPICS TO BE COVERED

1

Linear Inequalities

2

Solution of Linear inequalities

3

Related Problems

INTRODUCTION



Inequality:

* An algebraic expression with the sign of inequality.

- $<$ → less than
- $\leq$ → less or equal to.
- $>$ → Greater than
- $\geq$ → Greater or equal to.
- $a < b$ → a is less than b
- $a \leq b$ → a is less or equal to b .

• $a > b$ → a is greater than b

• $a \geq b$ → a is greater or equal to b .

Linear Inequality:

* Linear expression with the sign of inequality.

• $2x + 1 < 0$ (Linear inequality in one variable)

• $2x + y - 2 \geq 0$ (Linear inequality in two variables)

General form (In one variable)

- $ax + b < 0$
 - $ax + b > 0$
 - $ax + b \leq 0$
 - $ax + b \geq 0$
- where $a \neq 0$

Question: Which of the following is a linear inequality?

(a) $2x + x^2 > 0$ ✗

(b) $\sqrt{3}x + y^2 \leq 0$ ✗

(c) $\sqrt{3}x + 2 \leq 0$ ✓

(d) $x + 1 \geq x^2$ ✗

Question: Which of the following is a linear inequality?

~~(a)~~ $(x-1)(x+1) < x^2 - 2x$

(b) $x^2 + y^2 \geq 0$ ✗

(c) $(x+1)(x+2) < 2x^2$ ✗

(d) None of the above ✗

(c) $x^2 + 2x + x + 2 < 2x^2$
 $\Rightarrow x^2 + 3x + 2 - 2x^2 < 0$
 $\Rightarrow -x^2 + 3x + 2 < 0$

~~(a)~~ $x^2 - 1 < x^2 - 2x$
 $\Rightarrow -1 < -2x$
 $\Rightarrow -1 + 2x < 0$
 $\Rightarrow 2x - 1 < 0$

Basic Rules

$$3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$



- If $a > b$ and $b > c$, then $a > c$
- If $x > y$, then $x \pm c > y \pm c$ for any real number c
- * If $x > y$ and $a > 0$, then $ax > ay$.
- * If $x > y$ and $a < 0$, then $ax < ay$.
- If $x > y$ and x, y have the same sign, then $\frac{1}{x} < \frac{1}{y}$
- If $x > y \geq 0$, then $x^a > y^a$ for any $a \in \mathbb{R}^+$

- $3 > 2$ & $2 > 1 \Rightarrow 3 > 1$.
- $3 > 2 \Rightarrow 3 + 9 > 2 + 9$
& $3 - 9 > 2 - 9$
- $3 > 2 \Rightarrow 9 \times 3 > 9 \times 2$.
- $3 > 2 \Rightarrow -9 \times 3 < -9 \times 2$
- $3 > 2 \Rightarrow \frac{1}{3} < \frac{1}{2}$

$$3 > 2 \\ (3)^2 > (2)^2$$

$$3 > 2 \\ (3)^{-2} < (2)^{-2} \\ \frac{1}{9} < \frac{1}{4}$$

EXAMPLE

Solve $30x < 200$ when (i) x is a natural number (ii) x is an integer.

(Sol) Given that $30x < 200$

$$\Rightarrow \frac{\cancel{30}x}{\cancel{30}} < \frac{\cancel{200}}{\cancel{30}} = \frac{20}{3}$$

$$\Rightarrow x < \frac{20}{3}$$

Therefore (i) Since x is a natural number
So, the solution set = $\{1, 2, 3, 4, 5, 6\}$

(ii) Since x is an Integer, So
the solution set = $\{\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$

EXAMPLE

Solve $5x - 3 < 3x + 1$ when (i) x is an integer (ii) x is a real number.

(Sol) Given that $5x - 3 < 3x + 1$

$$\Rightarrow 5x - 3x < 1 + 3$$

$$\Rightarrow 2x < 4$$

$$\Rightarrow x < \frac{4}{2} = 2$$

$$\Rightarrow x < 2$$

(i) Since x is an integer. So,

the solution set = $\{\dots, -5, -4, -3, -2, -1, 0, 1\}$

(ii) Since x is a real number, so
the solution set = $(-\infty, 2)$



EXAMPLE

Solve $\frac{5-2x}{3} \leq \frac{x}{6} - 5$ Show the graph of the Solutions on the number line.

(Sol) Given that $\frac{5-2x}{3} \leq \frac{x}{6} - 5$

$$\Rightarrow \frac{5-2x}{3} \leq \frac{x-30}{6}$$

$$\Rightarrow \cancel{6} \left(\frac{5-2x}{\cancel{3}} \right) \leq x-30$$

$$\Rightarrow 10-4x \leq x-30$$

$$\Rightarrow -4x-x \leq -30-10$$

$$\Rightarrow -5x \leq -40$$

$$\Rightarrow x \geq \frac{-40}{-5}$$

$$\Rightarrow x \geq 8$$

Therefore the Solution Set = $[8, \infty)$



EXAMPLE

Solve $-5 \leq \frac{2-3x}{4} \leq 9$.

(Sol) Given that $-5 \leq \frac{2-3x}{4} \leq 9$.

Here, Consider the following!

$$-5 \leq \frac{2-3x}{4}$$

$$\Rightarrow -5 \times 4 \leq 2-3x$$

$$\Rightarrow -20 \leq 2-3x$$

$$\Rightarrow 3x \leq 2+20$$

$$\Rightarrow 3x \leq 22$$

$$\Rightarrow x \leq \frac{22}{3} \text{ --- (1)}$$

$$\text{Also, } \frac{2-3x}{4} \leq 9$$

$$\Rightarrow 2-3x \leq 9 \times 4$$

$$\Rightarrow 2-3x \leq 36$$

$$\Rightarrow -3x \leq 36-2$$

$$\Rightarrow -3x \leq 34$$

$$\Rightarrow x \geq -\frac{34}{3} \text{ --- (2)}$$

From (1) and (2), we get

$$-\frac{34}{3} \leq x \leq \frac{22}{3}$$

So, the solution set = $\left[-\frac{34}{3}, \frac{22}{3}\right]$.

Hence the result.

EXAMPLE

$$\text{Solve } \frac{x-2}{x+5} > 2.$$

(Sol) Given that $\frac{x-2}{x+5} > 2$

$$\Rightarrow \frac{x-2}{x+5} - 2 > 0$$

$$\Rightarrow \frac{x-2-2x-10}{x+5} > 0$$

$$\Rightarrow -\frac{x+12}{x+5} > 0$$

$$\Rightarrow -\frac{(x+12)}{x+5} > 0$$

$$\Rightarrow \frac{x+12}{x+5} < 0$$

Case 1st: $x+12 < 0$ & $x+5 > 0$

$$\Rightarrow x < -12 \text{ & } x > -5.$$

$\Rightarrow$ there is no such $x \in \mathbb{R}$ such that $x < -12$ & $x > -5$.

Case 2nd: $x+12 > 0$ & $x+5 < 0$

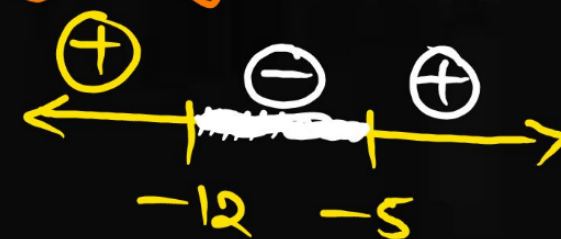
$$\Rightarrow x > -12 \text{ & } x < -5.$$

$$\Rightarrow -12 < x < -5 \Rightarrow x \in (-12, -5)$$

Therefore the required solution set
 $= (-12, -5) \cup \phi = (-12, -5).$

$$\frac{x+12}{x+5} < 0$$

By wavy Curve Method



$$x \in (-12, -5)$$

Solution set = $(-12, -5)$.

QUESTION

If $-3x + 17 < -13$, then

1 ✓ $x \in (10, \infty)$

2 ✗ $x \in [10, \infty)$

3 ✗ $x \in (-\infty, 10]$

4 ✗ $x \in [-10, 10)$

$$\begin{aligned} -3x + 17 &< -13 \\ \Rightarrow -3x &< -13 - 17 \\ \Rightarrow -3x &< -30 \\ \Rightarrow x &> \frac{-30}{-3} \\ \Rightarrow x &> 10 \\ \Rightarrow x &\in (10, \infty) \end{aligned}$$

QUESTION

Given that x, y and b are real numbers and $x < y, b < 0$, then

1 $\frac{x}{b} < \frac{y}{b}$ ✗

$$x < y$$

2 $\frac{x}{b} \leq \frac{y}{b}$ ✗

$$\frac{x}{b} > \frac{y}{b}, b < 0$$

3 $\frac{x}{b} > \frac{y}{b}$ ✓✓

$$\frac{x}{b} < \frac{y}{b}, b > 0$$

4 $\frac{x}{b} \geq \frac{y}{b}$ ✗

Remark

* $|x| \leq a \Rightarrow -a \leq x \leq a$

* $|x| < a \Rightarrow -a < x < a$

* $|x| \geq a \Rightarrow x \geq a \text{ or } x \leq -a$

* $|x| > a \Rightarrow x > a \text{ or } x < -a$

$\rightarrow (-\infty, -a] \cup [a, \infty)$

$\rightarrow (-\infty, -a) \cup (a, \infty)$

EXAMPLE

Solve $\sqrt{x-5} - \sqrt{9-x} > 1$, x is an integer.

Homework Problem.
#.

QUESTION

If x is a real number and $|x| < 3$, then

1 $x \geq 3$ ✗ $|x| < 3$

2 $-3 < x < 3$ ✓ $-3 < x < 3$

3 $x \leq -3$ ✗

4 $-3 \leq x \leq 3$ ✗

QUESTION

If $|x + 2| \leq 9$, then

- 1 $x \in (-7, 11)$ ✗
- 2 $x \in [-11, 7]$ ✓
- 3 $x \in (-\infty, -7) \cup (11, \infty)$ ✗
- 4 $x \in (-\infty, -7) \cup [11, \infty)$ ✗

(Sol) Given that $|x + 2| \leq 9$.

$$\Rightarrow -9 \leq x + 2 \leq 9$$

$$\Rightarrow -9 - 2 \leq \cancel{x + 2} - 2 \leq 9 - 2$$

$$\Rightarrow -11 \leq x \leq 7.$$

$$\Rightarrow x \in [-11, 7].$$

QUESTION

If $|x - 1| > 5$, then

1 $x \in (-4, 6)$ ✗

2 $x \in [-4, 6]$ ✗

3 $x \in (-\infty, -4) \cup (6, \infty)$ ✓

4 $x \in (-\infty, -4) \cup [6, \infty)$ ✗

$$\begin{aligned} * |x| > a \\ \Rightarrow x \in (-\infty, -a) \cup (a, \infty) \end{aligned}$$

$$\begin{aligned} \text{(Sol)} \quad & \text{Given that } |x-1| > 5 \\ \Rightarrow & x-1 > 5 \quad \text{or} \quad x-1 < -5. \\ \Rightarrow & x > 6 \quad \text{or} \quad x < -4 \\ \Rightarrow & x \in (-\infty, -4) \cup (6, \infty) \end{aligned}$$

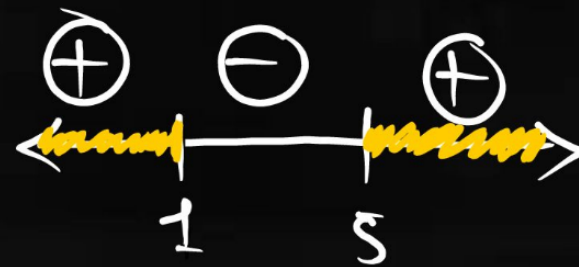
EXAMPLE

Solve the inequality : $(x - 1)(x - 5) \geq 0$, where x is a Real number

(Sol) Given that $(x - 1)(x - 5) \geq 0$.

By wavy-Curve Method, we get

$$x \in (-\infty, -1] \cup [5, \infty)$$



Therefore the required Solution set
 $= (-\infty, -1] \cup [5, \infty)$

QUESTION

The number of Integers satisfying the inequality $(x - 1)(x - 5) \leq 0$ is

1 3 ~~X~~

2 4 ~~X~~

3 5 ✓

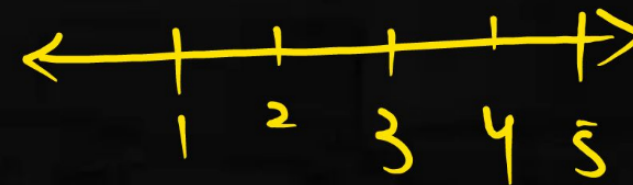
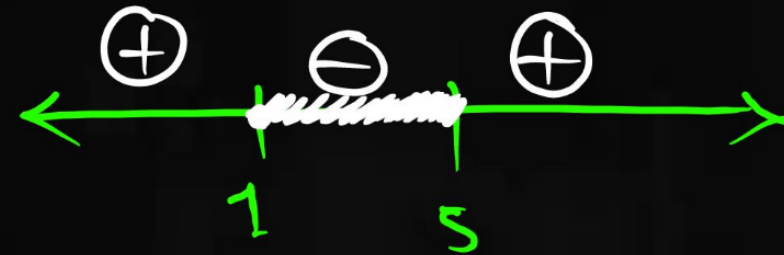
4 Infinite ~~X~~

(Sol) Given that $(x - 1)(x - 5) \leq 0$

By wavy - Curve Method

$$x \in [1, 5]$$

So, the Solution Set = $[1, 5]$



Thank
You